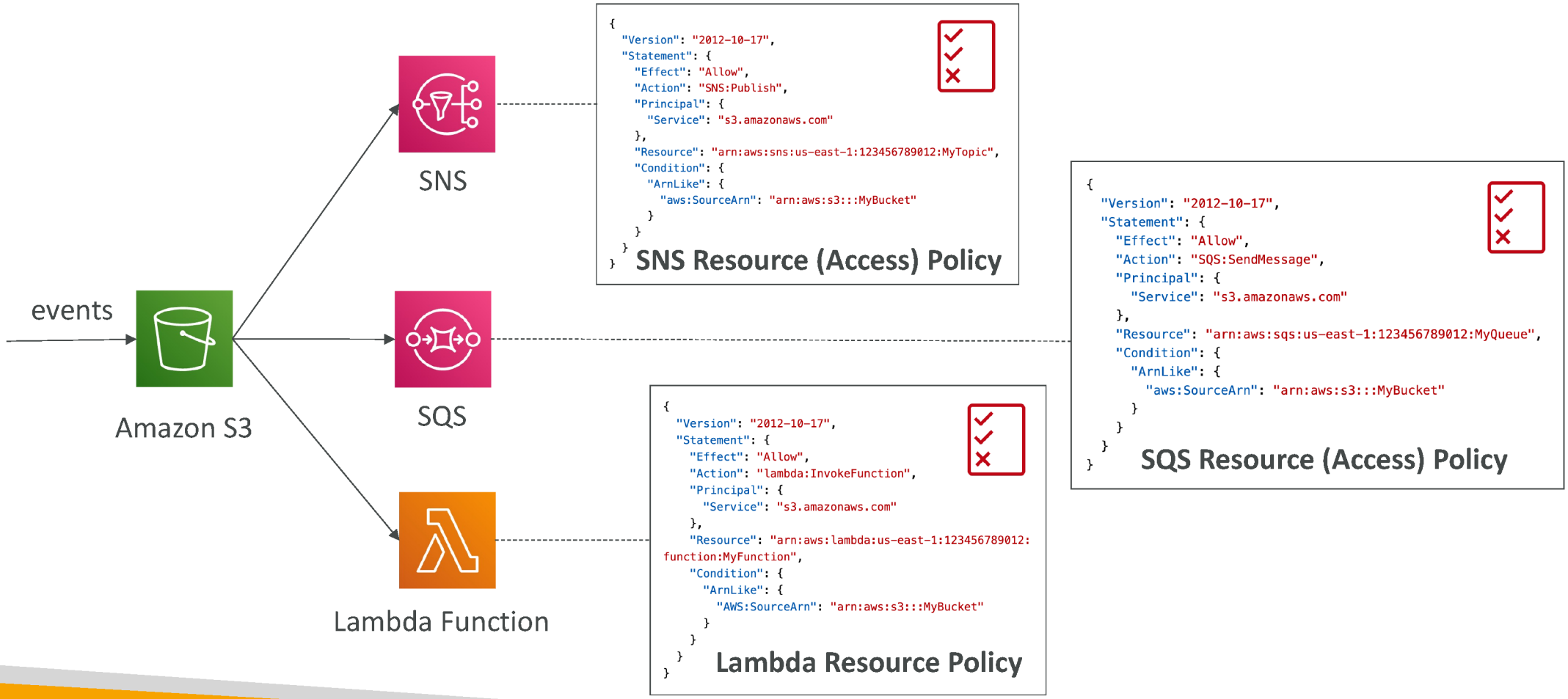
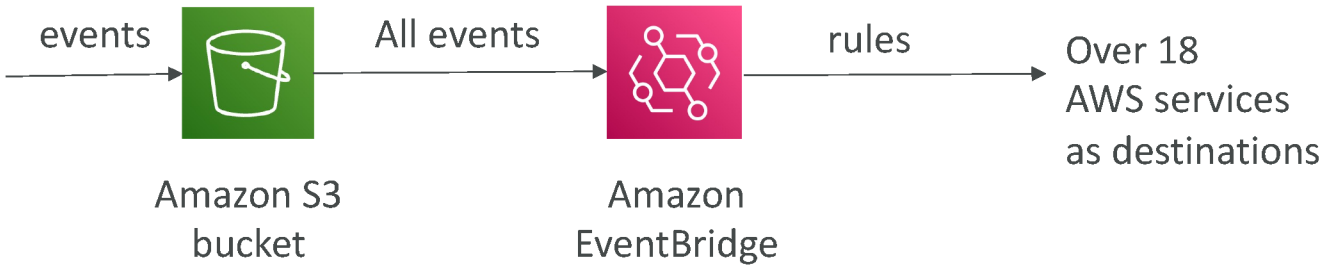
S3 Event Notifications – IAM Permissions



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S3 Event Notifications with Amazon EventBridge



- **Advanced filtering** options with JSON rules (metadata, object size, name...)
- **Multiple Destinations** – ex Step Functions, Kinesis Streams / Firehose…
- **EventBridge Capabilities** – Archive, Replay Events, Reliable delivery

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S3 – Baseline Performance

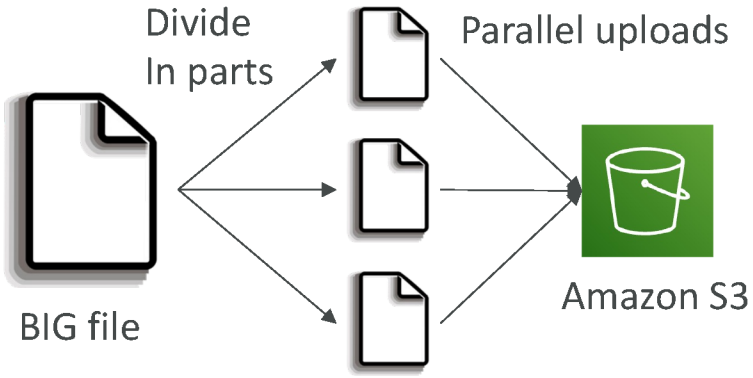
- Amazon S3 automatically scales to high request rates, latency 100-200 ms
- Your application can achieve at least **3,500 PUT/COPY/POST/DELETE or 5,500 GET/HEAD requests per second per prefix in a bucket.**
- There are no limits to the number of prefixes in a bucket.
- Example (object path => prefix):
  - bucket/folder1/sub1/file => /folder1/sub1/
  - bucket/folder1/sub2/file => /folder1/sub2/
  - bucket/1/file => /1/
  - bucket/2/file => /2/
- If you spread reads across all four prefixes evenly, you can achieve 22,000 requests per second for GET and HEAD

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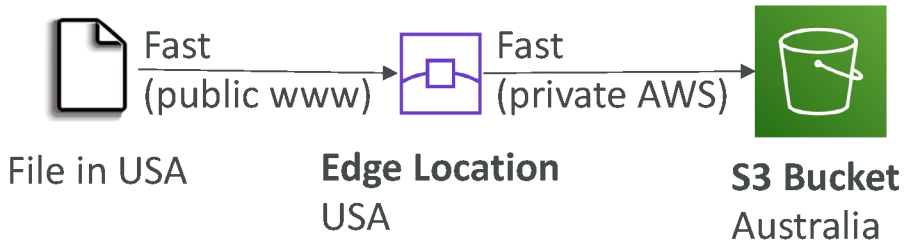
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S3 Performance

- **Multi-Part upload:**
  - recommended for files > 100MB, must use for files > 5GB
  - Can help parallelize uploads (speed up transfers)



- **S3 Transfer Acceleration**
  - Increase transfer speed by transferring file to an AWS edge location which will forward the data to the S3 bucket in the target region
  - Compatible with multi-part upload



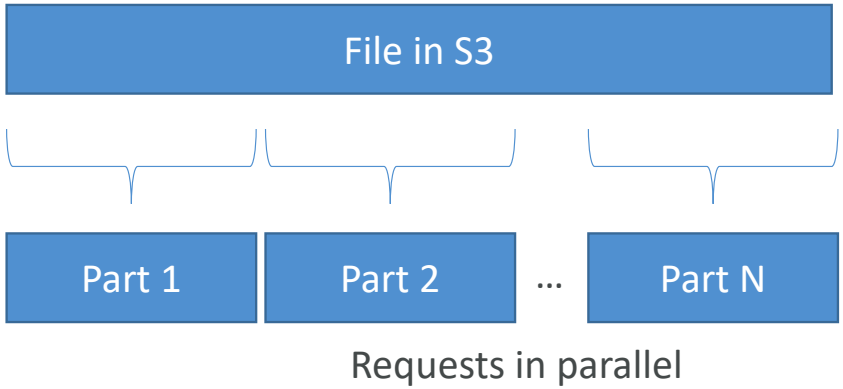
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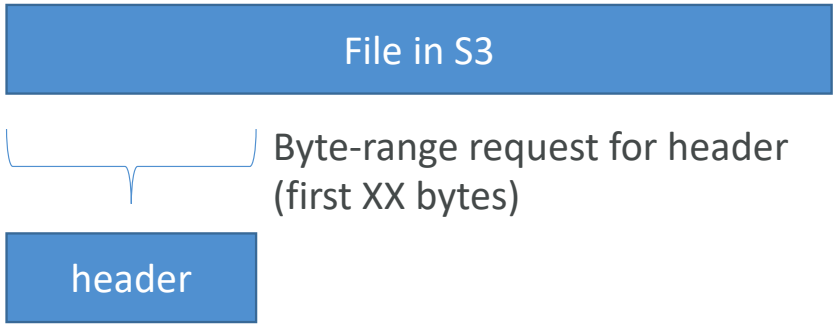
S3 Performance – S3 Byte-Range Fetches

- Parallelize GETs by requesting specific byte ranges
- Better resilience in case of failures

Can be used to speed up downloads



Can be used to retrieve only partial data (for example the head of a file)

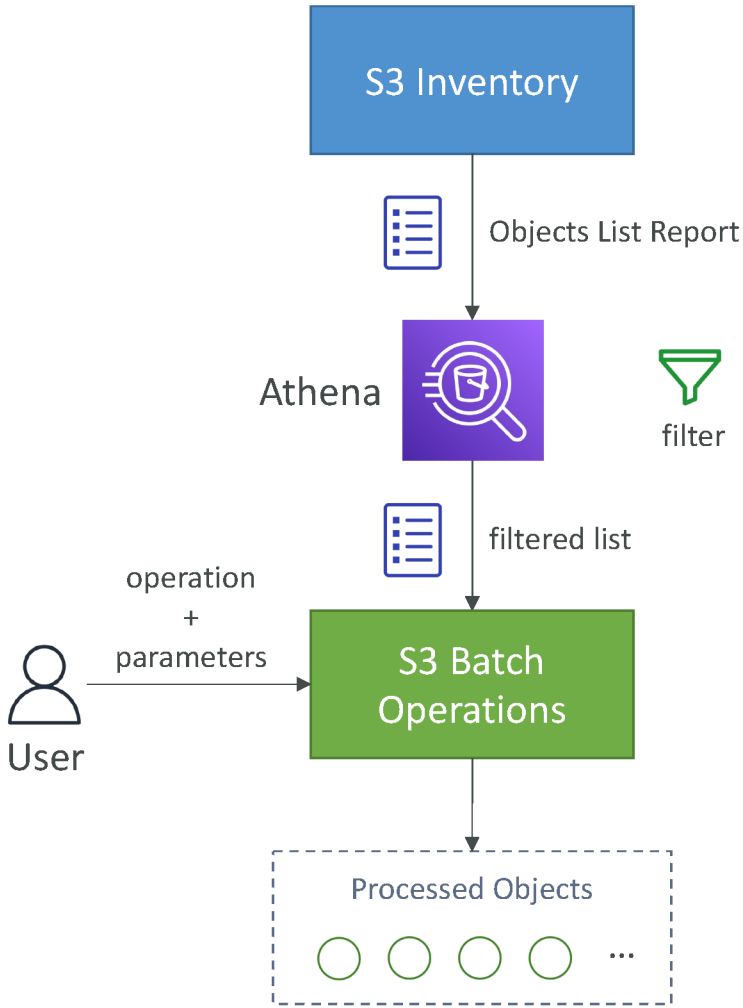


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S3 Batch Operations

- Perform bulk operations on existing S3 objects with a single request, example:
  - Modify object metadata & properties
  - Copy objects between S3 buckets
  - **Encrypt un-encrypted objects**
  - Modify ACLs, tags
  - Restore objects from S3 Glacier
  - Invoke Lambda function to perform custom action on each object
- A job consists of a list of objects, the action to perform, and optional parameters
- S3 Batch Operations manages retries, tracks progress, sends completion notifications, generate reports …
- **You can use S3 Inventory to get object list and use Athena to query and filter your objects**



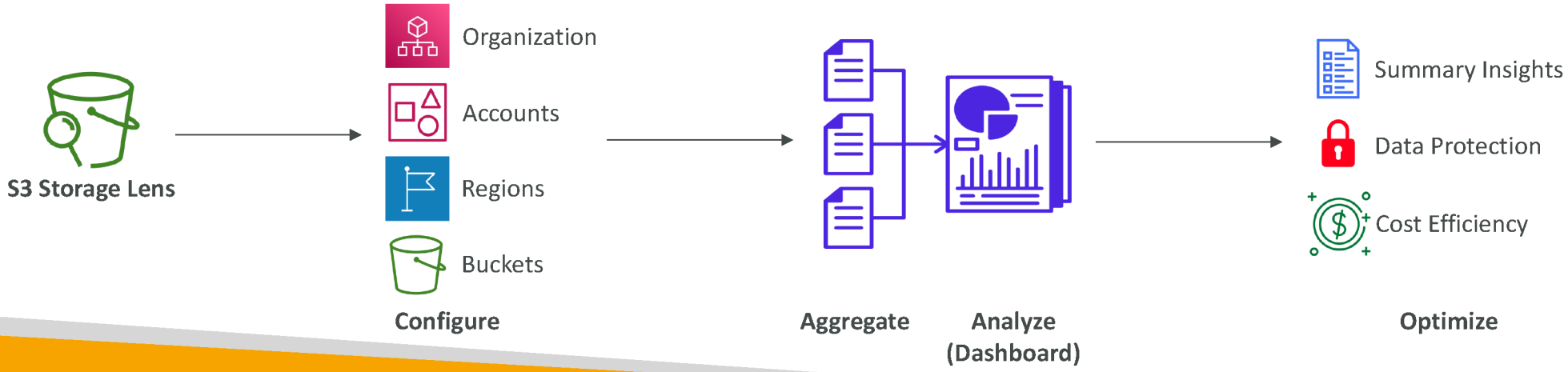
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S3 – Storage Lens



- Understand, analyze, and optimize storage across entire AWS Organization
- Discover anomalies, identify cost efficiencies, and apply data protection best practices across entire AWS Organization (30 days usage & activity metrics)
- Aggregate data for Organization, specific accounts, regions, buckets, or prefixes
- Default dashboard or create your own dashboards
- Can be configured to export metrics daily to an S3 bucket (CSV, Parquet)



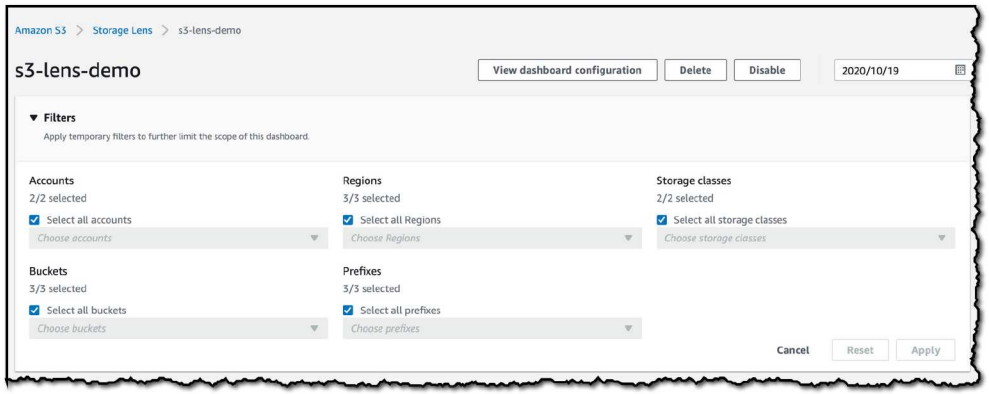
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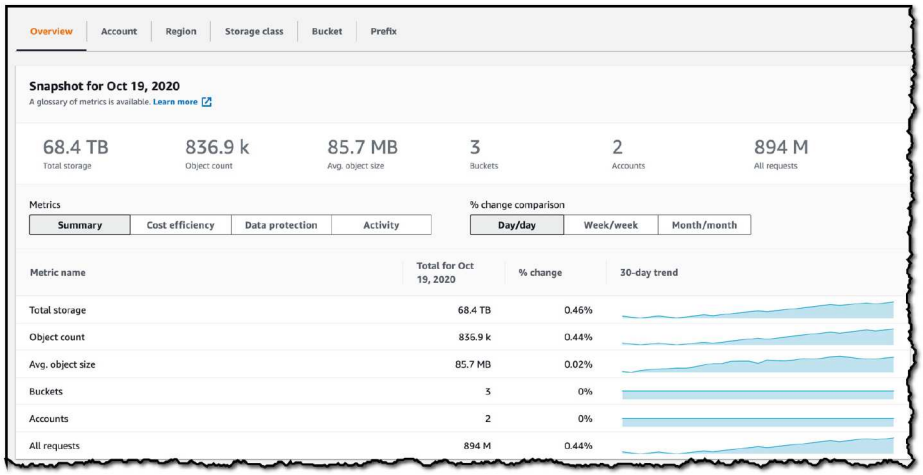
Storage Lens – Default Dashboard



- Visualize summarized insights and trends for both free and advanced metrics
- Default dashboard shows Multi-Region and Multi-Account data
- Preconfigured by Amazon S3
- Can't be deleted, but can be disabled



https://aws.amazon.com/blogs/aws/s3-storage-lens/



https://aws.amazon.com/blogs/aws/s3-storage-lens/

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Storage Lens – Metrics



- **Summary Metrics**
  - General insights about your S3 storage
  - StorageBytes, ObjectCount…
  - Use cases: identify the fastest-growing (or not used) buckets and prefixes

- **Cost-Optimization Metrics**
  - Provide insights to manage and optimize your storage costs
  - NonCurrentVersionStorageBytes, IncompleteMultipartUploadStorageBytes…
  - Use cases: identify buckets with incomplete multipart uploaded older than 7 days, Identify which objects could be transitioned to lower-cost storage class

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Storage Lens – Metrics



- **Data-Protection Metrics**
  - Provide insights for data protection features
  - VersioningEnabledBucketCount, MFADeleteEnabledBucketCount, SSEKMSEnabledBucketCount, CrossRegionReplicationRuleCount…
  - Use cases: identify buckets that aren't following data-protection best practices
- **Access-management Metrics**
  - Provide insights for S3 Object Ownership
  - ObjectOwnershipBucketOwnerEnforcedBucketCount…
  - Use cases: identify which Object Ownership settings your buckets use
- **Event Metrics**
  - Provide insights for S3 Event Notifications
  - EventNotificationEnabledBucketCount (identify which buckets have S3 Event Notifications configured)

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Storage Lens – Metrics



- **Performance Metrics**
  - Provide insights for S3 Transfer Acceleration
  - TransferAccelerationEnabledBucketCount (identify which buckets have S3 Transfer Acceleration enabled)
- **Activity Metrics**
  - Provide insights about how your storage is requested
  - AllRequests, GetRequests, PutRequests, ListRequests, BytesDownloaded…
- **Detailed Status Code Metrics**
  - Provide insights for HTTP status codes
  - 200OKStatusCount, 403ForbiddenErrorCount, 404NotFoundErrorCount…

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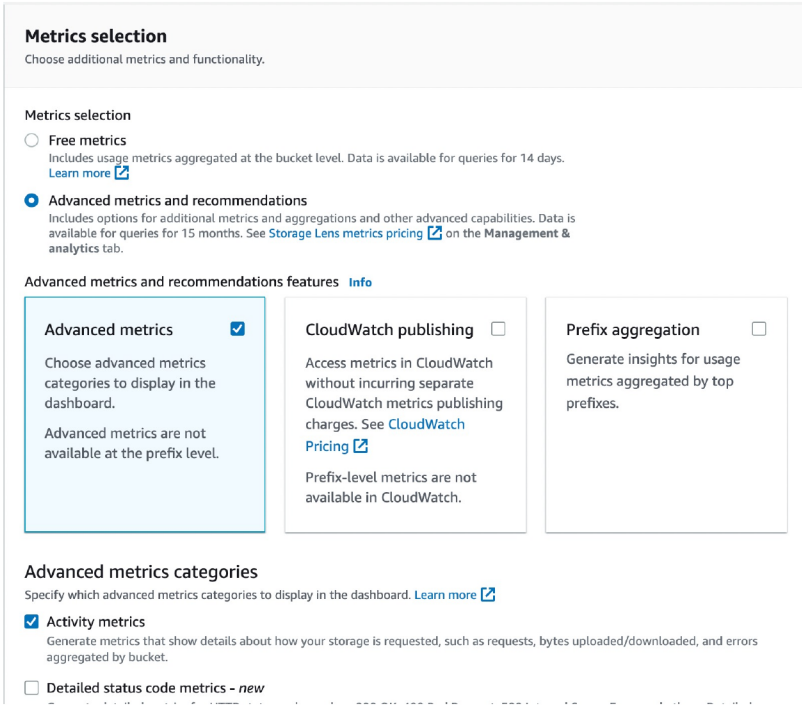
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Storage Lens – Free vs. Paid



- **Free Metrics**
  - Automatically available for all customers
  - Contains around 28 usage metrics
  - Data is available for queries for 14 days

- **Advanced Metrics and Recommendations**
  - Additional paid metrics and features
  - **Advanced Metrics** – Activity, Advanced Cost Optimization, Advanced Data Protection, Status Code
  - **CloudWatch Publishing** – Access metrics in CloudWatch without additional charges
  - **Prefix Aggregation** – Collect metrics at the prefix level
  - Data is available for queries for 15 months



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Amazon S3 – Security

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Amazon S3 – Object Encryption



- You can encrypt objects in S3 buckets using one of 4 methods

- **Server-Side Encryption (SSE)**
  - **Server-Side Encryption with Amazon S3-Managed Keys (SSE-S3)** – Enabled by Default
    - Encrypts S3 objects using keys handled, managed, and owned by AWS
  - **Server-Side Encryption with KMS Keys stored in AWS KMS (SSE-KMS)**
    - Leverage AWS Key Management Service (AWS KMS) to manage encryption keys
  - **Server-Side Encryption with Customer-Provided Keys (SSE-C)**
    - When you want to manage your own encryption keys
- **Client-Side Encryption**

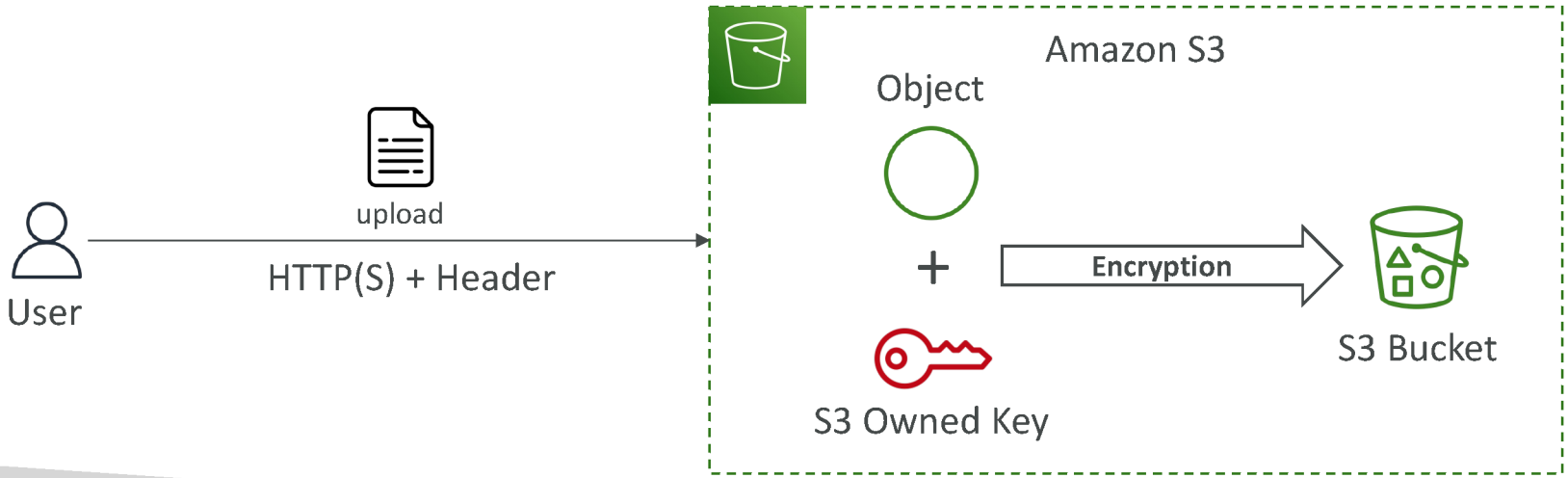
- It's important to understand which ones are for which situation for the exam

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Amazon S3 Encryption – SSE-S3

- Encryption using keys handled, managed, and owned by AWS
- Object is encrypted server-side
- Encryption type is **AES-256**
- Must set header **"x-amz-server-side-encryption": "AES256"**
- **Enabled by default for new buckets & new objects**

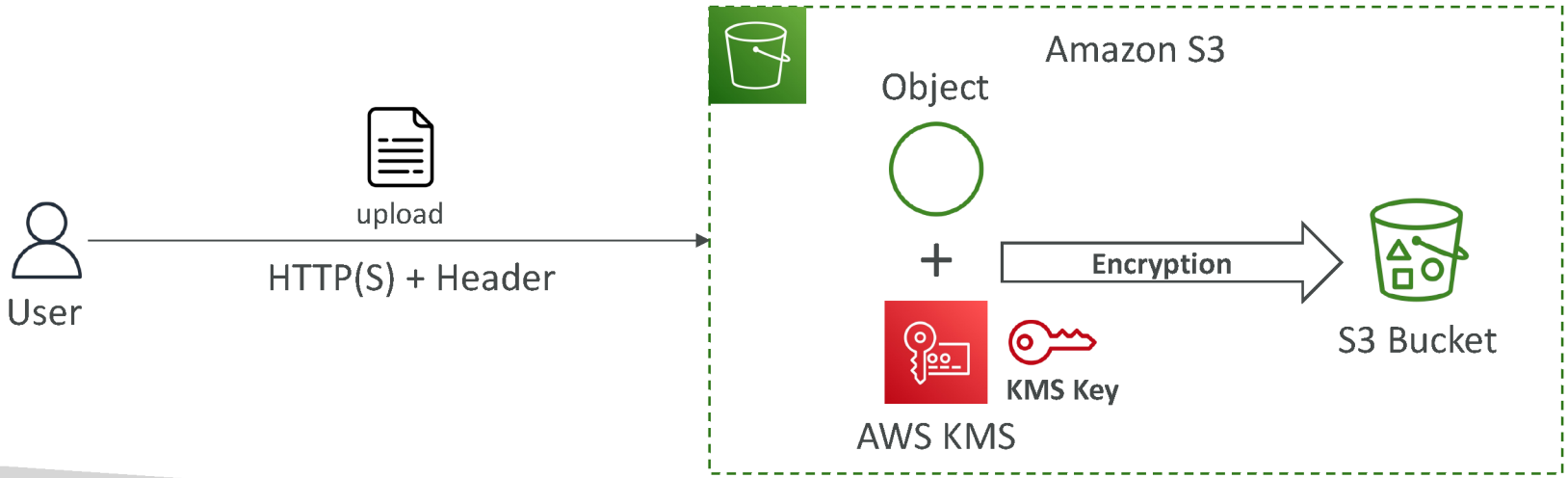


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Amazon S3 Encryption – SSE-KMS

- Encryption using keys handled and managed by AWS KMS (Key Management Service)
- KMS advantages: user control + audit key usage using CloudTrail
- Object is encrypted server side
- Must set header **"x-amz-server-side-encryption": "aws:kms"**

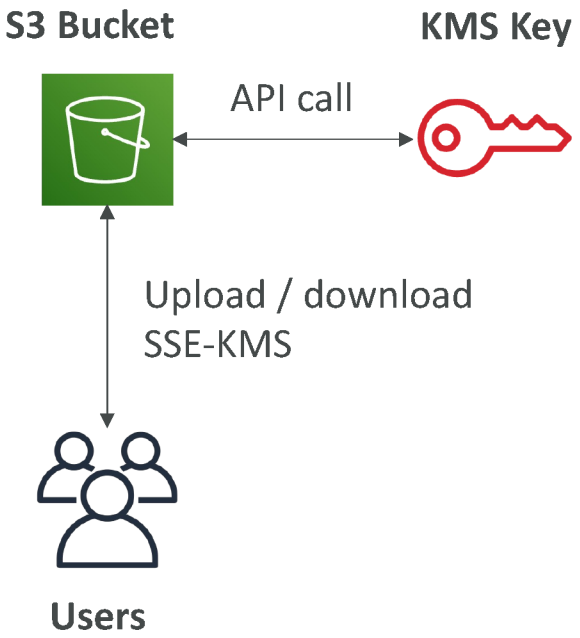


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SSE-KMS Limitation

- If you use SSE-KMS, you may be impacted by the KMS limits
- When you upload, it calls the **GenerateDataKey** KMS API
- When you download, it calls the **Decrypt** KMS API
- Count towards the KMS quota per second (5500, 10000, 30000 req/s based on region)
- You can request a quota increase using the Service Quotas Console

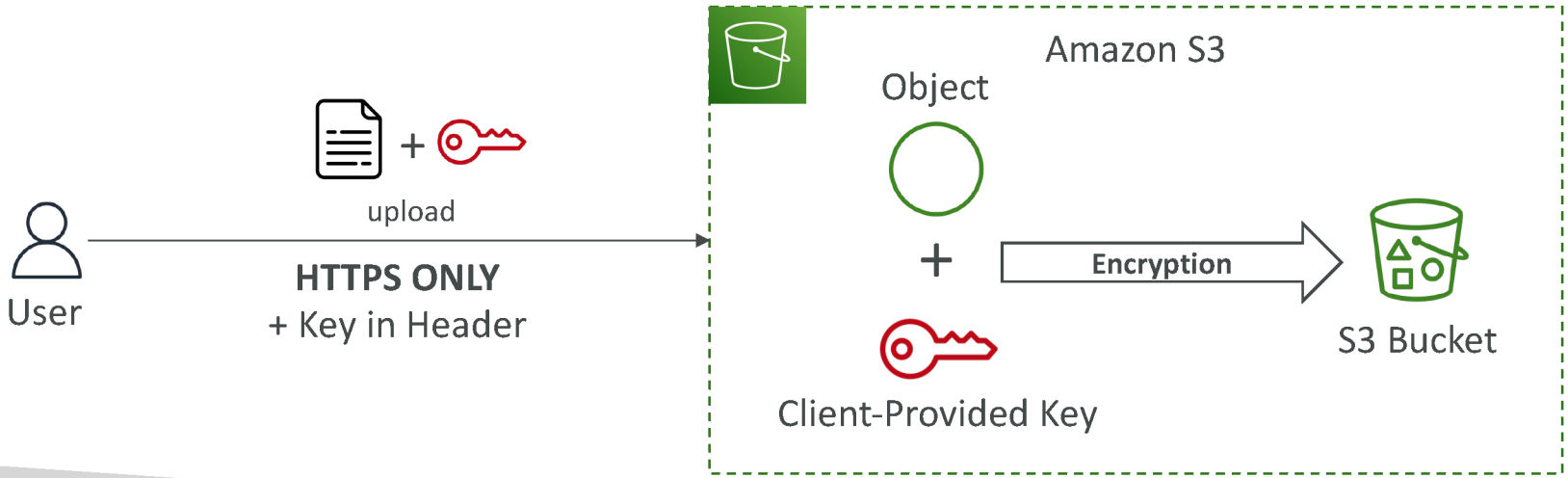


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Amazon S3 Encryption – SSE-C

- Server-Side Encryption using keys fully managed by the customer outside of AWS
- Amazon S3 does **NOT** store the encryption key you provide
- **HTTPS must be used**
- Encryption key must provided in HTTP headers, for every HTTP request made

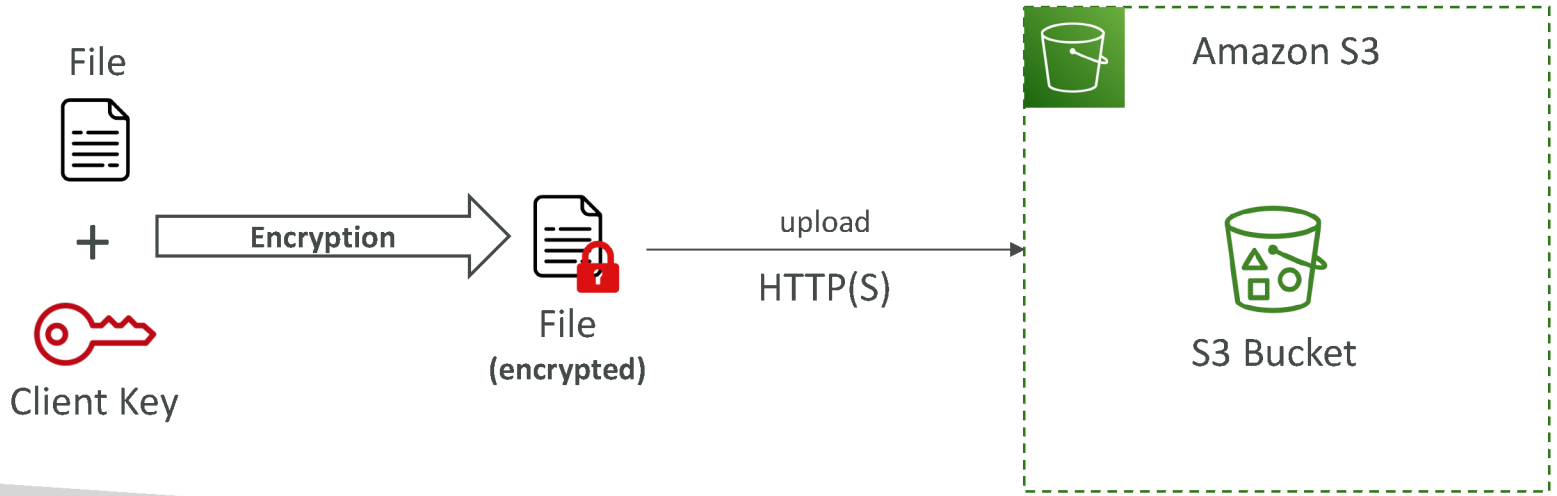


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Amazon S3 Encryption – Client-Side Encryption

- Use client libraries such as **Amazon S3 Client-Side Encryption Library**
- Clients must encrypt data themselves before sending to Amazon S3
- Clients must decrypt data themselves when retrieving from Amazon S3
- Customer fully manages the keys and encryption cycle



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Amazon S3 – Encryption in transit (SSL/TLS)

- Encryption in flight is also called SSL/TLS

- Amazon S3 exposes two endpoints:
  - **HTTP Endpoint** – non encrypted
  - **HTTPS Endpoint** – encryption in flight

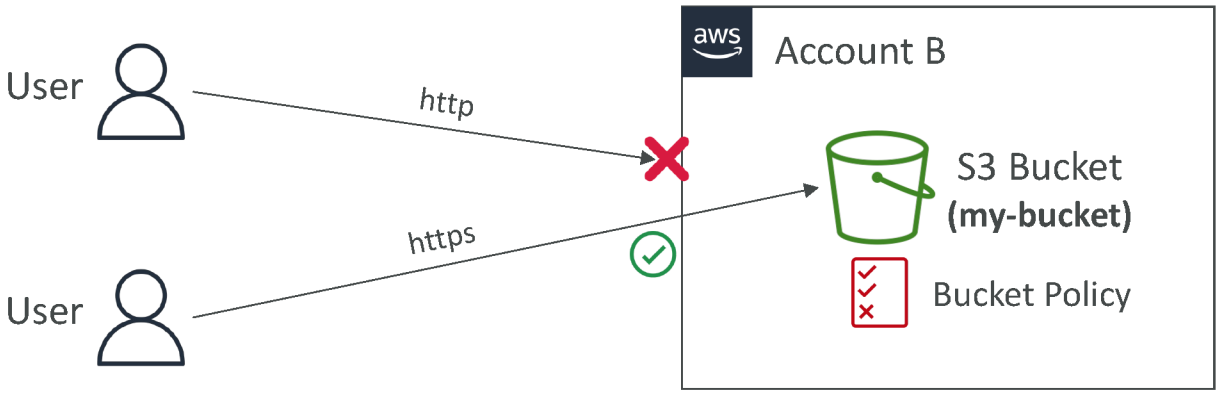


- **HTTPS is recommended**
- **HTTPS is mandatory for SSE-C**
- Most clients would use the HTTPS endpoint by default

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Amazon S3 – Force Encryption in Transit aws:SecureTransport




```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Deny",
            "Principal": "*",
            "Action": "s3:GetObject",
            "Resource": "arn:aws:s3:::my-bucket/*",
            "Condition": {
                "Bool": {
                    "aws:SecureTransport": "false"
                }
            }
        }
    ]
}
```

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